

Pre-processing

- Resize 224 × 224
- ImageNet Normalisation
- No Crop, No Augmentation

Same image → three independent raters

3 Annotators & Label + Confidence 0–100

- μ_H mean
- σ_H spread
- H_label entropy

σ_H, H_label used by Module B (§5.2)

**Input & Preprocessing**

## ResNet-50 Branch



| Conv 7×7, 64, s2 | Max Pool 3×3, s2 | Stage 1 [1×1,3×3,1×1] ×3 | Stage 2 × 4 | Stage 3 × 6 | Stage 4 × 3 | Global Avg Pool | Dropout p = 0.3 | FC 10 classes |
|---|---|---|---|---|---|---|---|---|
| $112^2 \times 64$ | $56^2 \times 64$ | $56^2 \times 256$ | $28^2 \times 512$ | $14^2 \times 1024$ | $7^2 \times 2048$ | 2048 | 2048 | 10 |

## EfficientNet-B4 Branch



| Conv 3×3, 48, s2 | MBConv1 k3 × 2 | MBConv6 k3 × 4 | MBConv6 k5 × 4 | MBConv6 k3 × 6 | MBConv6 k5 × 8 | Conv 1×1 + GAP | Dropout p = 0.3 | FC 10 classes |
|---|---|---|---|---|---|---|---|---|
| $112^2 \times 48$ | $112^2 \times 24$ | $56^2 \times 32$ | $28^2 \times 56$ | $14^2 \times 160$ | $7^2 \times 448$ | 1792 | 1792 | 10 |

**Ensemble Fusion**

Softmax $p^R$ → Softmax $p^E$ → $0.5\,p^R + 0.5\,p^E$ → 10-dim class distribution → Loss: CMP + Human-Informed CE ($\lambda = 0.5$)

**Dual-Backbone Feature Extraction**

Temp. Scaling T

ECE 0.147 → 0.009

RAPS Conformal Sets
r = 0.077, uninformative

→ Module B feature (§5.2)

MC Dropout
T = 50 passes

Dropout retained at inference

Variance → Epistemic

→ Module B feature (§5.2)

μ_H (x) = 0.38

mean of three annotators

p_AI (x) = 0.97

HACM (x)=
p_AI(x) − μ_H(x)

0.97 − 0.38 = + 0.59

Instance-Level Misalignment
(illustrative extreme; dataset mean |HACM| =0.134)

Dataset Aggregate
mean | HACM |

**Uncertainty, Calibration & Metric**